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Exam Class !

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Part IV: Answer the following questions based on the instruction and write your answer ONLY on the space provided as concise and clear as possible. **(13 pts.)**

1. **Fill in the blank space (3 pts.)**

   A. ________________ is process in which sequence of nucleotides in DNA converting or copying into sequence of nucleotides in mRNA.

   B. ________________ is a coenzyme and common electron carrier in animal cells which stored energy in the form of electrons that can be appointed to make ATP during cellular respiration.

   C. ________________ is the most important biological enzyme in photosynthesis which converts inorganic CO_2 into organic molecules that can be used by the cell.

2. **Compare aerobic respiration and anaerobic respiration. By describing O_2 requirement, which part of cell it occurs, and maximum amount of net ATP produced per glucose?** ------- **(3 pts.)**

	Aerobic Respiration	Anaerobic Respiration
Occurs in:		
Oxygen:		
Net ATP		

3. **Write two reasons why Gregor Mendel choose garden peas (*Pisum sativum*) for his hybridization experiment?** -- **(2 pts.)**

4. ***Write at list four risk factors of metabolic disorders.*** ------------------------ **(2 pts.)**

5. ***Define the following biological words*** -- **(3 Pts.)**

   A. *Photorespiration*

   B. *Chemoautotrophs*

   C. *Gluconeogenesis*

10. Which one of the following mechanism of evolution proposed by Charles Darwin mechanism of natural selection Consists of: -

A. Use and Disuse
B. Inheritance Acquired Traits
C. Natural Selection
D. 'A' and 'B' are correct

11. ________ is CO_2 acceptor in the Calvin cycle of photosynthesis.

A. RuBP (Ribulose 1, 5-Bisphosphates)
B. NADPH (Nicotinamide Adenine Dinucleotide Phosphate)
C. NADH (Nicotinamide Adenine Dinucleotide)
D. Rubisco or RuBP Carboxylase

12. Which of the following is the key intermediate compound linking glycolysis to the Kreb's cycle in cellular respiration?

A. Citric Acid B. Acetyl CoA C. Pyruvic Acid D. Adenosine Triphosphate

13. A cross that is expected to produce 50 % homozygous and 50 % heterozygous is /are?

A. TT x Tt B. Tt x Tt C. tt x Tt D. all of the above

14. In the Dark Reactions of photosynthesis

A. CO_2 is incorporated into an organic molecule
B. Oxygen is produced
C. Water is split
D. ATP is synthesized

15. Which of the following is an example Chromosomal Mutation

A. Substitution B. Addition C. Deletions D. Euploidy

16. Which molecule is an electron donor in oxygenic photosynthesis?

A. O_2 B. CO_2 C. H_2S D. H_2O

17. Where in chloroplast does the Light Independent Reaction of photosynthesis takes place?

A. Stroma B. Grana C. Thylakoid D. 'B' and 'C' are correct

18. Which one of the following is an example of C_4 plants?

A. Cactus B. Orchids C. Corn D. Pineapple

2. The part of **tRNA** molecule that matches with the sequence of nucleotides on **mRNA** molecule during protein synthesis is called ________.

   A. Codon B. Anticodon C. Transcodon D. Amino Acid

3. A woman with **Blood Type A** has a child with **Blood Type O**. Which of the following blood type **COULD NOT** be the possible father's blood type of this child?

   A. Type A B. Type B C. Type AB D. Type O

4. If twenty percent (20%) of the nucleotide bases in human DNA are Thymine (T), what will be the percentage of Cytosine (C)?

   A. 20% B. 30% C. 40% D. 70%

5. Which one of the following statement about enzymes is **FALSE**?

   A. Enzymes modify the rate of chemical reactions.
   B. Enzymes lower the activation energy for the reaction they catalyze.
   C. Enzymes alter the equilibrium of the reaction they catalyze.
   D. Enzymes are effective in extremely small quantities.

6. ________ is the protein portion of a conjugate enzyme?

   A. Apoenzymes B. Cofactors C. Holoenzyme D. Coenzymes

7. Which one of the following classes of enzyme catalyze the formation of carbon and other bonds using energy from ATP?

   A. Transferases B. Lyases C. Ligases D. Hydrolases

8. Substance that binds with enzyme to block E-S complex formation is called________.

   A. Activator B. Inhibitor C. Substrate D. Cofactor

9. Which one of the following stage of cellular respiration takes place in mitochondria matrix of the cell?

   A. Glycolysis
   B. Formation of Acetyl CoA
   C. Electron Transport Chain
   D. Krebs Cycle

PART I: Write TRUE if the statement is correct or FALSE if the statement is incorrect. **(1pt each)**

1. As long as ATP and NADPH are available, formation of organic molecule from atmospheric CO_2 photosynthetic process may occur in the absence of light.
2. *Chlorophyll a* is primary pigment of all plants, blue-green algae, and cyanobacteria that participate directly in the light reaction of photosynthesis.
3. In Mendel's classic pea cross, dominant trait will be express in phenotype only if a plant is homozygous for that trait.
4. An-oxygenic photosynthesis is a commonly known photosynthetic process seen in plants, algae and cyanobacteria.
5. Activation Energy is the maximum amount of energy, which is required to start a chemical reaction.

Part II: Match the Genetic Terms (column "B") with their description in column "A" and write your answer in capital letter on the space provided. **(1 pt. each)**

Column "A"	Column "B"
1. Alternative version of a gene	A. Recessive allele
2. Having two different alleles for a gene	B. Dominant allele
3. Has no effect on phenotype in a heterozygous	C. Phenotype
4. Heritable feature that varies among individuals	D. Genotype
5. Genetic makeup of an organisms	E. Alleles
	F. Homozygous
	G. Heterozygous

Part III: Choose the correct answer from the given alternatives and write on the space provided. **(1.5 pts. each)**

1. If the sequence of nucleotides in template stand of a DNA is **-TTAGCCTT-**, what will be the sequence of nucleotides in mRNA after transcription?

   A. **-AATCGGAA-** B. **-AAUCGGAA-** C. **-AAGGCUAA-** D. **-UUAGCCUU-**

16. Write two factors which affect the rate of diffusion of molecules across the plasma membrane using carrier mediated transport?

17. ________________ class of enzymes catalyze reactions involving formation of carbon and other bonds with energy from ATP.

18. ____________________ is the sum of cellular activities that breakdown large molecules to release energy and synthesize building materials from smaller ones.

19. Enzymes speed up chemical reactions by decreasing the ____________ energy of reactions.

20. The light-dependent reactions of photosynthesis are important because they produce ____________ and ____________ that are used as raw materials for the light-dependent reactions.

________________ END ________________

33. Which of the following is least likely to cross a cellular membrane by simple diffusion?

A. carbon dioxide B. potassium ion C. oxygen D. steroid hormone

34. Categories of enzymes that have two components which are a protein and a cofactor are called

A. Simple enzymes

B. Conjugate enzymes

C. Isozymes

D. Metallo-enzymes

35. Increasing the substrate concentration increases the reaction rate. But this effect stops at some point and reactions proceed at a constant rate beyond this point. This is due to

A. Reduced efficiency of the enzyme

B. Competition between substrate molecules

C. Saturation of all available enzymes

D. Reduced kinetic energy of substrate molecules

36. Glycolysis occurs in which part of the cell?

A. Inner membrane of the mitochondria

B. Matrix of the mitochondria

C. Cytoplasm

D. Ribosome

37. Organisms that acquire energy through photosynthesis are called

A. Photoautotrophs

B. Chemoautotrophs

C. Heterotrophs

D. Saprotrophs

38. Which of the following occurs during the Krebs cycle?

A. FADH2 accepts two electrons in order to form FAD.

B. NAD+ is reduced to form NADH.

C. Pyruvate is converted into acetyl-CoA.

D. Acetyl-CoA combines with oxaloacetate to release NADH

23. A biological molecule of the plasma membrane which plays essential role in maintaining fluidity, flexibility and stability of the lipid membrane is

A. cholesterol
B. carbohydrate
C. cytoskeleton
D. Na+-K+ pump

24. If the function of a cellular organelle is sorting, modification, and packaging of synthesized proteins, then this organelle must be a

A. Smooth ER B. Ribosome C. Rough ER D. Golgi body

25. Which of the following organelles have their own DNA?

A. Lysosomes and peroxisomes
B. Smooth end Rough ER
C. Mitochndria and chloroplast
D. Golgi body and ribosomes

26. What are the two main parts of the plasma membrane?

A. Phospholipids and cholesterol
B. Proteins and carbohydrates
C. Carbohydrates and cholesterol
D. Phospholipids and proteins

27. The internal skeleton of a cell is composed of

A. Microfilaments, microtubules, and intermediate filaments
B. Intermediate filaments and cellulose
C. Centrioles, microtubules, and cellulose
D. Microfilaments and centrioles

28. A form of diffusion in which there is a net movement of water down its concentration gradient through a selectively permeable membrane is called

A. Simple diffusion B. Facilitated diffusion C. Active transport D. Osmosis

29. Which of the following statements is correct?

A. Facilitated diffusion uses proteins to transport molecules across the cell membrane
B. In active transport, substances move from an area that is more concentrated to less concentrated
C. In passive transport, substances move from an area that is less concentrated to more concentrated
D. Passive transport requires energy while active transport does not

30. Which of the following transport processes across the plasma membrane are correctly matched with the mechanisms of transport? Movement of

A. Water along concentration gradient ------- passive diffusion
B. Na+ along concentration gradient ---------- active transport
C. Water along concentration gradient -------- facilitated diffusion
D. Na+ against concentration gradient -------- passive diffusion

31. Which transport mechanism is used to transport charged and polar molecules along a concentration gradient?

A. Passive diffusion
B. Active transport
C. Bulk transport
D. Facilitated diffusion

32. A cell containing sodium ions at a concentration of 0.05 mM was kept in a 0.005 mM of sodium solution. If more sodium ions happen to enter into the cell, the mechanism used must have been

A. diffusion B. osmosis C. active transport D. exocytosis

45. What are the possible causes of metabolic disorders?

A. Age

B. Family history of genetic metabolic disorder

C. Obesity and lack of exercise

D. All

Short Answer

1. ______________________ is a process for experimentation which is used to explore observations and give answers to research questions.
2. List three major characteristics of living things
3. Write down three steps used in the scientific method.
4. What are the three types of RNA involved in protein synthesis?
5. Mention the three components of a nucleotide.
6. What do you call fatty acids that have at least one double bond in their hydro-carbon chain?
7. Based on their solubility, vitamins are classified as ____________________ and ______________
8. Write two functions that minerals have in living organisms.
9. Mention two physiological and structural functions of lipids in living cells.
10. Monosaccharides that have the aldehyde functional group are classified as ____________________
11. ____________________ is part of the nucleus where ribosomes are synthesized.
12. What are the two unique features of mitochondria?
13. ____________________ is a kind of enzyme inhibition in which the inhibitor binds on a site other than the active site.
14. What do you call the mechanism of transport of polar molecules or ions according to a concentration gradient (downhill) using carrier molecules but with no expenditure of energy?
15. What is endocytosis?

39. What is the name of the glycolytic enzyme that converts dihydroxyacetone phosphate (DHAP) to lyceraldehydes-3-phosphate (GAP)?

A. Hexokinase
B. Trios phosphate isomerase
C. Aldolase
D. Enolase

40. Which of the following occurs in both photosynthesis and cellular respiration?

A. Calvin cycle
B. Chemiosmosis
C. Glycolysis
D. Krebs cycle

41. Where exactly in the cell does the electron transport chain occurs?

A. Cytoplasm
B. Inner membrane of the mitochondria
C. Matrix of the mitochondria
D. Ribosome

42. The synthesis of ATP using the process of chemiosmosis takes place during which phase of respiration?

A. Glycolysis B. TCA cycle C. Electron transport chain D. Biosynthesis

43. NADPH2 and ATP produced by the light-dependent reactions are used to reduce

A. Carbon Monoxide
B. Hydrogen
C. Carbon Dioxide
D. Oxygen

44. What is the fluid-filled space of chloroplasts that contains enzymes for the light-independent reactions of photosynthesis?

A. Thylakoid
B. Grana
C. Matrix
D. Stroma

C 10. Among the following alternatives one is **not** the importance of lipids.

A. Used as the main components of cell membrane.

B. Serve for protection and cellular communication.

C. Catalyzing chemical reactions.

D. Insulation of heat and water.

E. All of the above

III. Match words or phrases from column "B" to those under "A" by writing the letter of your choice. (5 pts).

Column A	Column B
F 1. Inorganic biological molecule	A. Glucose
H 2. Golgi body	B. Triglycerides
D 3. Structural polysaccharides	C. Controls all the activities of the cell
C 4. Nucleus	D. Cellulose
A 5. Monosaccharide	E. Synthesize energy rich compound ATP
	F. Water
	G. Glycogen
	H. Modifies, sorts and packs materials synthesized in the cell

IV. **Answer the following questions by filling correct terms or phrase in the blank spaces.** *(2 points)*

1. The three major parts of a typical animal cell are Mito Chondria, Lysosomes and Nucleus/cell membrane (1.5 pts)

2. *From the amino group (–NH_2), carboxyl group (–COOH), and R group of the amino acid, the group that vary in the 20 different amino acids is the* protein *(0.5 pt).*

R group

17.5/20

A 6. An organelle containing powerful hydrolytic enzymes capable of digesting & removing unwanted cellular foreign materials and serve as the intracellular digestive system is:

A. Lysosomes B. Vacuole C. Ribosome D. Golgi bodies E. Endoplasmic reticulum

D 7. Which one of the following is **not true** ?

A. All plasma membranes are made up of lipids and proteins and small amount of carbohydrates.
B. Depending on their roles & functions within organisms, cells differ in their size, shape, and internal structure.
C. Lipid-soluble substances can passively diffuse through the plasma membrane down their electro-chemical gradients.
D. Smooth endoplasmic reticulum is used for the production of proteins.
E. The movement of molecules in active transport is only in one direction.

C 8. Which of the following is **true**?

A. Fatty acids possessing one or more double bonds are saturated fatty acids.
B. In research, experimental group **does not** receive treatment or experimental manipulation.
C. Starch and glycogen are the main energy storage carbohydrates in plants & animals, respectively.
D. In prokaryotes, mitochondria are the power houses of a cell.
E. All

B 9. Which one of the following is **not** among the factors that influences the rate of net diffusion of a substance across a biological membrane?

A. Permeability of the membrane
B. Presence of energy
C. Distance through which diffusion must take place
D. Molecular weight of the substance
E. Surface area of the membrane

General Biology (Biol 1012) Mid-term Exam (for regular 1st year students) April 15, 2021

17.5+6 = 23.5

25

ARBA MINCH UNIVERSITY
COLLEGE OF NATURAL SCIENCES
DEPARTMENT OF BIOLOGY

ID No [redacted] section C

Name [redacted]

Time allowed 75 minutes, Total Mark: 25%

This examination contains 31 questions in 5 parts, use only block letters for parts two and three

Part one Write true if the statement is true or false if the statement is false (0.5 pt.)

1. True A theory can be disproved after repeated experiments.
2. False The theory of eternity of life states that life originated as a result of gradual process.
3. False The nucleoli, the chromatin and the nuclear envelope are the major parts of the nucleus.
4. True Hydrolysis reaction consumes water and releases energy.
5. False Glucose and fructose are non-reducing sugars.
6. True Mitochondria have the ability to replicate themselves even when the cell to which they belong is not undergoing cell division.
7. False Vitamins are one of important biomolecules required by the body in large quantities.
8. False The vacuole is an important organelle that maintains turgor pressure in animal cells.
9. False Osmosis occurs towards the direction of low solute concentration.
10. False Adenine and thymine are purines.

Part two: Match the functions under column "B" to organelles under column "A" (0.5 pt.)

A

1. D Lysosomes
2. F Golgi complex
3. H Mitochondria
4. C Nucleus
5. E
6. I

B

A. Glucose synthesis
B. Detoxification
C. Cellular control
D. Cellular digestion
E. Protein synthesis
F. Protein packaging
G. Waste storage

INFINIX
AI CAMERA

20. ______ The α-carbon of amino acids does *not* bond with
   A. the amino group
   B. the carboxylic group
   C. a hydrocarbon chain
   D. a hydroxyl group
21. ______ Which of the following is/are not monosaccharides?
   A. ribose
   B. fructose
   C. sucrose
   D. galactose
22. ______ Combination of a sugar, a nitrogenous base and a phosphate group is termed as
   A. histone
   B. amino acid
   C. nucleic acid
   D. nucleotide
23. ______ A bond joining nitrogenous bases of nucleic acids is
   A. Glycosidic bond
   B. Peptide bond
   C. Hydrogen bond
   D. Carboxylic bond
24. ______ Level of protein structure in which amino acid chain formed is folded into a unique 3-dimensional shape called α-helix or β-pleated sheets is
   A. Quaternary structure
   B. Secondary structure
   C. Tertiary structure
   D. Primary structure
25. ______ Which of the following is **false** about Plasma/cell membrane?
   A. selective permeability
   B. made up of lipids and nucleic acids
   C. have polar hydrophilic head
   D. have non-polar hydro
26. ______ Assume you put a peeled potato into a beaker of solution containing a very high salt concentration. After a day, what will happen to the potato?
   A. it will gain mass
   B. it will remain the same
   C. it will lose mass
   D. it will gain water

Part four: Fill the following blank spaces with appropriate words (1 pt.)

27. Lipids that have at least one double bond between carbon atoms in the tail chain and can accommodate at least one more hydrogen atom are ______________________.
28. Complex macromolecules that store and transmit genetic information are ______________.
29. Series of enfolding of inner mitochondrial membrane that contain proteins __________.

Part five: Give short answers for the following questions (2 pts.)

30. Write a statement of the cell theory that disproves the theory of spontaneous generation.

31. Mention four functions of proteins.

Part I: Write true if the statement is correct else false on the space provided. (1 point each)

[redacted] 1. Emerging technology is not continuing development of existing technology.

[redacted] 2. Networking is connecting computers or other electronic devices together so that they share files or resources.
device - device   data - data (logic)

[redacted] 3. Structured data have a predefined data model.
3

[redacted] 4. Industrial revolution three is also known as Technological evolution
digital

Part II: Choose the right answer from the alternatives on provided Space (1 point for each)

[redacted] 5. What are the main components of Data value Chain in data Science?
- A. Data processing
- B. Data management
- C. Data access
- D. Data Analysis

- Aquisition
- analysis
- curation
- storage
- usage

management
- access
- process
- storage

[redacted] 6. Which of the following is correct about evolution and revolution?
- A. Evolutionary - incremental and takes place gradually, over time.
- B. Evolutionary - incremental in that it happens step by step, little by little.
- C. revolutionary change is deep and Change is fundamental, dramatic, and often irreversible
- D. All are correct

[redacted] 7. The final or main cause of the Industrial Revolution was the effects created by ____
- A. Agricultural Revolution
- B. rise of digital technology
- C. machine learning
- D. Smart & autonomous systems fueled by data

[redacted] 8. In which Industrial Revolution (IR) increasing use of steam power and rise of the factory system
- A. IR 1.0
- B. IR 2.0
- C. IR 3.0
- D. IR 4.0

8

[redacted] 9. ______ is the re-structuring or re-ordering of data by people or machines to increase their usefulness and add values for a particular purpose
- A. Data value chain
- B. Data curation
- C. Data Processing Cycle
- D. Data Acquisition

[redacted] 10. Which data type in computer programming perspective is used to store real numbers
- A. Integer(int)
- B. Boolean(bool)
- C. Floating-point number(float)
- D. Alphanumeric string(string)

[redacted] 11. Which of the following is not character of big data?
- A. Volume
- B. Velocity
- C. Veracity
- D. None

usage)

[redacted] 12. ______________ refers to the communication and interaction between a human and a machine via a user interface.
- A. Human machine interaction
- B. Programmable devices
- C. Human computer interface
- D. A&C

HMI

13. ______________ is the process of gathering, filtering, and cleaning data before it put in a data warehouse.
- A. Data acquisition
- B. Data curation
- C. Data access
- D. Data usage

Selale university

Salale University
College of Natural Sciences
Department of Chemistry

5. Which one of the following is **not** an intensive property?
   A. Temperature
   B. Density
   C. Heat
   D. Pressure
6. Which one of the following statement is **not true** about colligative properties of a solution?
   A. Vapor pressure of a solution is decreases as concentration of solute increases.
   B. Boiling point of a solution is increases as concentration of solute increases.
   C. Osmotic pressure of a solution is decreases as concentration of solute increases.
   D. Freezing point of a solution is decreases as concentration of solute increases.
7. How many Significant figures in this number 0.003048?
   A. 7 B. 6 C. 4 D. 5
8. Which one of the following is **true**?
   A. If both components of a solution are non-volatile, the vapor pressure of the solution is the sum of the individual partial pressures.
   B. Colligative properties are properties that depend on the number of solute particles in solution and on the nature of the solute particles.
   C. The solubility of gases in water decreases with increasing temperature.
   D. When solute and solvent molecules mix to form a solution, there is decrease in randomness.
9. From the following ways of expressing concentration, one is used for more diluted solution?
   A. Molarity B. Molality C. ppm D. ppb
10. Which one of the following is **not** both accurate and precise?
    A. Determining the purity of gold: 99.9999%, 99.9998%, 99.9998%, 99.9999%
    B. Measurements for the height of the 50 cm plant: 49.7 cm, 50.0 cm, 49.8 cm, 50.1 cm, 50.2 cm
    C. Testing the volume of a batch of 25-mL pipettes: 27.02 mL, 26.99 mL, 26.97 mL, 27.01 mL
    D. A and B

Part I. Write "*TRUE*" if the statement is correct and "*FALSE*" if the statement is incorrect on the answer sheet (1 pts each).

1. Osmotic pressure is a selective passage of solute from dilute to concentrated solution. T
2. A hypothesis always leads to the formulation of a law.
3. **Captive zeros** result from measurement are always non-significant.
4. A solution which contains a nonvolatile solute needs a higher temperature to reach its vapor pressure than the boiling point of the pure solvent.
5. The theoretical **van't Hoff factor** *i* for the electrolyte solution is equal to 1. F

Part II. Choose the correct answer from the given alternatives and write your answer on answer sheet by capital letter (1 pts each)

1. Which one of the following is **incorrect** about factor that affects solubility?
   A. Both solute and solvent have similar molecular structure and polarity.
   B. The solubility of most solid increases as temperature increase.
   C. Gases are more soluble at low temperature.
   D. As Pressure increase the solubility of the gas decrease.
2. __________ cannot be separated into other substances by **physical processes**, but can be decomposed into other substances by **chemical processes**.
   A. Heterogeneous mixture
   B. Homogeneous mixture
   C. Element
   D. Compound
3. A commercial "**berekina**" solution contains 5 % sodium hypochlorite, NaOCl. What is the mass of NaOCl in a bottle containing 2500 g of bleaching solution?
   A. 125 g B. 100 g C. 150 g D. 50 g
4. Which one of the following significant digits is obeys rules for rounding numbers?
   A. 0.028675 rounds -to 0.0287
   B. 18.3384 rounds - to 18.3
   C. 6.8752 rounds -to 6.88
   D. All

5%

$\frac{}{100} \times 5\% = 41$

e. Ag_2CO_3 Argon Carbonate f. H_3PO_4 Hydrogen Phosphate

Part IV Work out /Show the necessary steps for the following questions

4. The organic compound ethanol in the beer boils at 351.52° K in aqueous solutions. What is the temperature required to boil ethanol in degree Celsius and degree Fahrenheit? (2pts)

Given Required Solution

$T_K = $ $K_c = 351.52$ C = ? F = ?

$T_C = K - 273.15$
$= 351.52 - 273.15$
$= 78.37 C$

$T_F = \frac{9}{5}(T_C) + 32$
$= \frac{9 \times 78}{5} + 32$
$= \frac{357}{5} + 32$
$T_F = \frac{357 + 160}{5}$

5. The chemical symbol of chromium is $^{52}_{24}Cr$. What are the number of protons, electrons and neutrons of Cr^{3+} and Cr^{6+}? (2pts)

P = 24 = Z

P = Z = n un charged election elements but his is charged

then A = P + n
n = A - P
= 52 - 24 = 28, e = A - P = 28 - 24 = 4

P = 24
e = 4
n = 28

6. Element X has an elemental atomic mass of 7 amu and has two naturally occurring isotopes, ^6X and ^7X. Their masses are 6 amu and 8 amu respectively. What are the natural abundances of the isotopes of element X? (2 pts)

Oxygen $^{12}_{2}O$ and $^{16}_{2}O$

6 O

Part I- Write true if the statement is correct and false if the statement is incorrect (1 pts each).

1. A tentative explanation of observations is known as theory.
2. Chemical change not always produces matter that differ from the matter present before the change.
3. If the supporting data is available, the law will be modified.
4. All chemical properties are intensive properties.
5. Blood is an example of heterogeneous mixture.
6. Matter in the mixture can also have properties of more than one physical state.

Part-II Choose the correct answer for the following questions from the given alternatives (1 pts each)

1. Which alternative determines the size of an atom?
   A. Number of electron
   B. Nucleus
   C. Atomic number
   D. Neutron
2. Which one is **not** macroscopic domain of Chemistry
   A. Melting of matter
   B. Solubility
   C. Color change
   D. Shape of molecule
3. Aqueous solution of NaCl is considered as ____________.
   A. Compound
   B. Homogeneous mixture
   C. Heterogeneous mixture
   D. Element
4. The number of significant figures in 45.0000 and 0.000045 are ____ and ____, respectively.
   A. 2 and 2
   B. 2 and 6
   C. 6 and 6
   D. 6 and 2
5. One of the following values is not equal to 1kg of pure water?
   A. 1 L of pure water
   B. $1dm^3$ of pure water
   C. $1000\ cm^3$ of pure water
   D. 1000g of beer
6. Which factor **doesn't** contribute for the uncertainty of measurements?
   A. The person doing the measurement
   B. The measuring device
   C. The environment where the measurement is being made

B. 0.75 D. None

Part III. Briefly explain the following essay items

1. Chemistry as central science shares knowledge with different disciplines of Science. So list some of the disciplines and the knowledge transfer from Chemistry to others? (1.5 pts)

-Chemistry is have demands this demands

- Chemistry is the central science of other

macroscope

macroscope and

symbolic

- to study wider

=> to study the nature of matter composition, properties and reactions. Then

2. Hydrogen is unique element in the periodic table. What properties of hydrogen resemble to group 1A and Group 7A elements? (1pts)

- Hydrogen is higher element

- hydrogen have two isotopes

hydrogen react with oxygen to form water molecules

3. Write the name of the following compounds based on IUPAC naming system?(1.5 pts)

a. Li_3N Limun Nitrode

b. $(NH_4)_2PO_4$ Nitrogen phosphat

c. $NaCl$ Nedium chloride

d. $FeCl_3.6H_2O$ iron chlolt

D. The nature of the material being measured E. None

7. Which of the following is not true

A. Precise measurements have less random error

B. Accurate measurements have low systematic error

C. All non-zero digits are significant

D. The more the number of significant you have the more error you made.

8. From the following measurements which one gives exact result without any uncertainty?

A. Number of students in the class C. Height of students in the class

B. Weight of students in the class D. B & C

9. Which one is **true** about Isotopes?

A. They have the same number of electron C. They have the same atomic mass

B. They have same number of neutron D. They have different number of proton

10. One of the following properties is different from the others?

A. Color C. electrical conductivity

B. Hardness, melting D. Rusting E. None

11. Among the following one results a chemical change.

A. Condensation of steam C. Melting of ice

B. Freezing of water D. Splitting of water molecule

12. Which of the following is correct when 18.85 is rounded to one decimal place?

A. 18 B. 18.8 C. 18.9 D. 18.85

N.B: Use the following information to answer question number 13

i. Testing these hypotheses by further experiments.

ii. Seeking patterns in the observations,

iii. Making observations,

iv. Formulating hypothesis to explain the observations,

v. Collecting information/data

13. As a Science student if you faced a problem of acidic soil in your land, which order of the scientific method do you follow to solve the problem?

A. i, ii,iii,iv,v C. iii,v,ii,i,v

B. iii,v,ii,iv,i D. v,iv,iii,ii,i

14. A piece of silver metal weighing 194.3g is placed in a graduated cylinder containing 242ml of water. The volume now reads 260.5ml. What is the density of the metal in gram per ml?

A. 0.802 C. 10.503

January-25/2022

Bahir Dar University College of Science
Department of Chemistry
Mid-Exam: General Chemistry (Chem. 1012)

Time allowed:- 80 min.

Constants: atomic number: Fe = 26, Al = 13, I = 53, S = 16, V = 23, Cr = 24, Mn = 25
Atomic mass: H=1, N=14, O=16, Na=23, P=31, Fe=56, Hg=201 g/mol

$N_A = 6.02x10^{23} mol^{-1}$

Multiple Choice: Place the letter of the correct answer on the separate sheet. (10 pts)

1. Which of the following is not homogenous mixture?
   A. salt & water B. 70% Ethanol C. Milk D. Air
2. Which of the following is extensive physical property of substance?
   A. Mass B. Color C. Melting point D. Temperature
3. Which of the following molecule is not diatomic molecule?
   A. Argon B. Oxygen C. Nitrogen D. Chlorine
4. The species that contains 24 protons, 26 neutrons and 22 electrons would be represented by the symbol:
   A. $^{50}V^{3+}$ B. $^{26}Cr^{2+}$ C. $^{50}Cr^{2+}$ D. $^{50}Mn^{2+}$

   24 + 26 = 50 22 $^{50}X^{+2}$
5. Element 'X' has two isotopes with 1:1 ratio with atomic number of 5. One isotope has a mass number of 10 and the other isotope has a mass number of 11. What is the average atomic mass of 'X'?
   A. 10.5 amu B. 21.0 amu C. 10.0 amu D. 11.0 amu

   0.5 × 10 + 0.5 × 11
6. Which one of the following is not an oxyacid?
   A. Nitric acid B. Sulfuric acid C. hydrochloric acid D. acetic acid

   5 + 5.5
   10.5 amu
7. What is the molarity of phosphoric acid in a solution labeled 20.0% phosphoric acid (H_3PO_4) by weight with a density of 1.12 g/mL? (At. mass: H=1, O=16 & P=31 g/mol)
   A. 0.98 M B. 2.3 M C. 2.7 M D. 3.6 M

   3 + 64 + 31 = 98
   $\frac{20 \times 1.12 \times 10}{98}$
8. Which of the following is an example of monatomic anion?
   A. NH_4^+ B. Mg^{2+} C. OH^{-1} D. S^{2-}
9. Which formula/name pair is incorrect?
   A. $Fe_2(SO_3)_3$: Iron (III) sulfite
   B. $FeSO_4$: Iron (II) sulfate
   C. FeS : Iron (II) sulfide
   D. $Fe_2(SO_4)_3$: Iron (III) sulfide

   200 × 1.12 1.12 2.2
   1.12 20
   006 98) 224
   000 196
   224 280
   400 196
   98
   294
10. How many fluorine atoms are there in a mole of uranium hexafluoride (UF_6)?
    A. 6.02×10^{23} B. 3.612×10^{22} C. 3.612×10^{24} D. 6.02×10^{24}

Mobile Phones are **NOT** Allowed.

$6 \times 6 \times 10^{23}$ 3.6 6.022 × 6 = 32 6.02 × 6 = 36.12 × 10^{23}

True/False: Read each statement below carefully. Write true if you think a statement is correct. Write False if you think the statement is incorrect. (3 pts)

T 1. Atoms of a matter are more compacted in liquid state than in gaseous state.
F 2. The first step of scientific studying process is making experiment.
F 3. The number of neutrons determines the identity of a chemical element.
F 4. Combining hydrogen and oxygen to make water is a physical process.
T 5. Number of moles of solute in equal volume of concentrated and diluted solution is equal.
6. Abebe prepared a NaOH solution by dissolving 8.0 g of NaOH in 100.0 mL of solution. From this solution, Abebe can prepare a 100 mL of 2.5 M NaOH solution by dilution.

Short Answer (2 pts each)

1. The weight of a millimole of $(NH_4)_3PO_4$ is: 0.149 g / 149×10^{-3} g
2. What volume of water should be added to 200 ml of a 0.01 M NaOH solution in order to change the concentration to 0.005 M NaOH? 400 mL
3. Give the number of protons and electrons in each of the following common ions:

Ions	# proton	# electron
Al^{3+}	13	10
I^-	53	54

$m_1V_1 = V_2M_2$

$\frac{0.01(200)ml}{0.005M} = V_2$

4. Fill the blanks in the following table.

Cation	Anion	Formula	Name
Ba^{+}	Cl^-	BaCl	Barium chloride
Co^{2+}	PO_4^{3-}	$CO_3(PO_4)_2$	cobalt (II) posphide

Work Out (Show the necessary steps clearly) (9 pts)

1. A 50 g sample of industrial wastewater was determined to contain 0.32 mg of mercury. What is the concentration (in ppm) of mercury in the wastewater?
2. Lithium has an elemental atomic mass of 6.941 amu and has two naturally occurring isotopes, ^{6}Li and ^{7}Li. Their masses are 6.02 amu and 7.02 amu respectively. What are the natural abundances of the isotopes of Lithium?
3. A 40.0 mL volume of 1.80 M $Fe(NO_3)_3$ is mixed with 20.0 mL of 0.8 M $Fe(NO_3)_3$ solution. Calculate the molar concentration of the final solution.

$N = 3\times14 = 42$

$O = 4\times16 = 64$

149

$m = mole \times M$
$= 1\times10^{-3}\times149$
$= 149\times10^{-3}$ / 0.149 g

$\frac{\text{mass in solute}}{\text{mass in solution}}\times10^6$

Name__________________

ID________

Time Allowed: 40'

Maximum weight 15%

Wachemo University

College of Natural & Computational Science

Department of Chemistry

Organic Chemistry test-I (chem1012)

<u>For 1st year Medicine Regular students</u>

I/ Write "true" if the statement is correct and "false" if it is incorrect. (0.5pt each)

1. Any compound with asymmetric center is optically active.

2. A molecule is Chiral if its two mirror image forms are superimposable.

II/ Choose the correct answer from the given alternatives. (0.5pt each)

1. Which of the following compound has asymmetric center?

A/ 2-butanol C/ 2-bromopropane

B/ 3-bromopentane D/ Ethanol E/ All have asymmetric center.

2. Which one of the following statement describes the formation of pi (π) bonds in carbon? It is formed by the :-

A/ sideways overlap of unhybridized p-orbitals.

B/ internuclear axis overlap of p-orbital.

C/ sideways overlap of an s & p-orbitals.

D/ sideways overlap of either s or p-orbitals.

E/ sideways overlap of hybridized p-orbitals.

3. In which of the following combination of orbitals hybridization <u>will</u> occur?

A/ 3s with 2p B/ 2s, 3p & 3d C/ 3s, 3p & 3d D/ 2s, 2p & 3d E/ none

4. Which one of the following statement is true about orbitals?

A/ In sp^2 hybridization of C unhybridized 2p-orbital is lower in energy than atomic 2p-orbitals.

B/ In sp^2 hybridization of C unhybridized 2p-orbital is lower in energy than atomic 2s-orbital.

C/ All hybridized and unhybridized orbitals are degenerate.

D/ Unhybridized 2p orbitals form a δ-bond through internuclear axis overlap.

E/ none of the above

5. Which one of the following statement **is true** about optical activity of molecules?

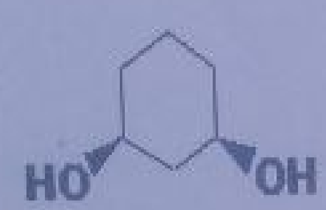
A/ Molecules with internal plane of symmetry are optically active.

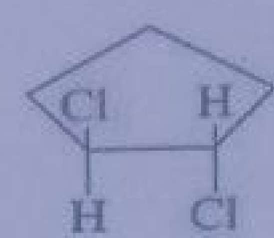
B/ Meso compounds are optically active.

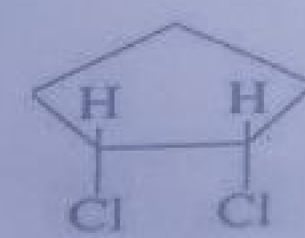
C/ Racemic mixture of two enantiomers is optically active.

D/ Molecules with one Chirality center are optically active.

6. Which of the following molecule **has no** internal plane of symmetry?

A/ 

B/ 

C/ 

D/ None

7. How many stereogenic (chirality) center the molecule given below has?

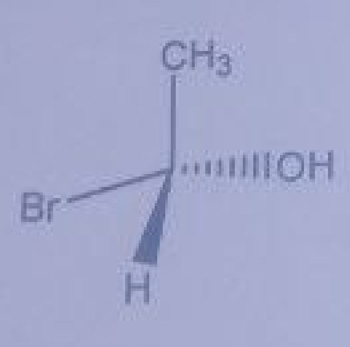
$CH_3CHCHCH_3$
Cl OH
3-chloro-2-butanol

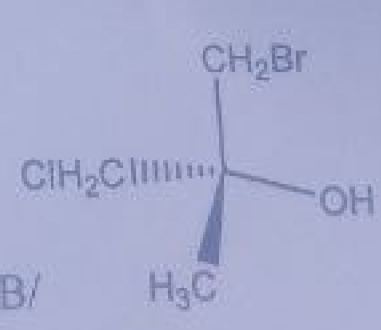
A/ 1
B/ 3
C/ 2
D/ 4 E/ it has no chirality

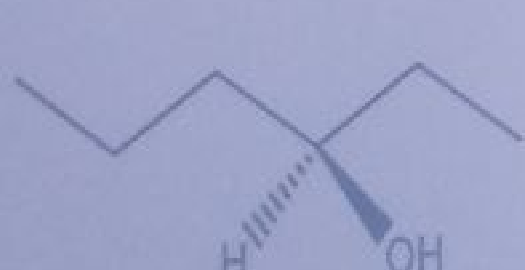
III. Answer the following questions accordingly

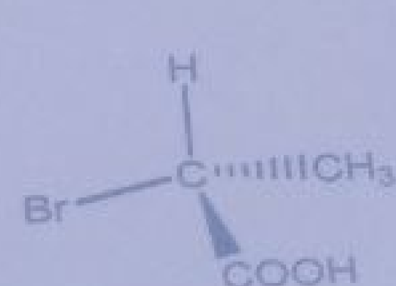
1. Label the following molecules structure as "S" or "R" configuration. **(2pt each)**

(Given, Z for each atoms H = 1, C = 6, O = 8, Br = 35 and Cl = 17)

A/ 

B/ 

C/ 

D/ 

2. Explain why anti-conformation of butane about C_2-C_3 rotation is more stable than other staggered and eclipsed form of conformations. (**1.5pt**)

3. Write some common and different physical properties of two enantiomers (**0.5pt**)

4. Draw and label the *E* and *Z* isomers for the compound given below (**1pt**)

$$CH_3CH_2\underset{\displaystyle Cl}{\underset{|}{C}}{=}CHCH_2CH_3$$

E- isomers

Z- isomers

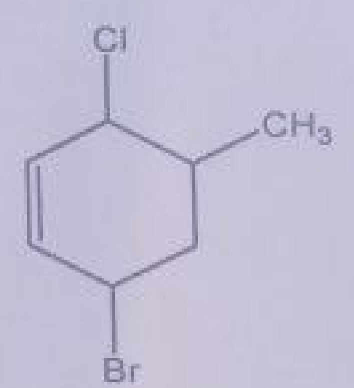
5. Write the *Z* & *E* isomers of 2-bromo-3-chloro-2-pentene (**1.5pt**)

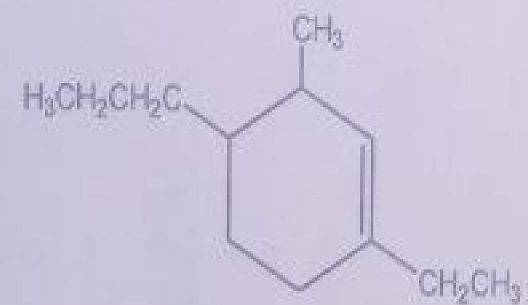
Z–isomers

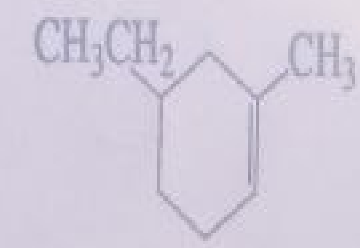
E-isomers

B/

6. Give systematic name for the organic molecule given below (**0.5pt each**):-

A/ 

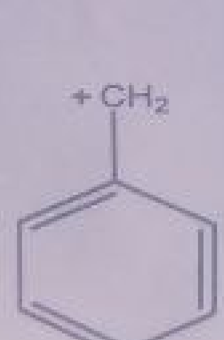
B 

C. 

7. Arrange the following organic acids in the increasing order of their inductive effect. (**1pts**)

$H_3C-\overset{\overset{O}{\|}}{C}-OH$ A , $ClH_2C-\overset{\overset{O}{\|}}{C}-OH$ B , $FH_2C-\overset{\overset{O}{\|}}{C}-OH$ C , $IH_2C-\overset{\overset{O}{\|}}{C}-OH$ D , $BrH_2C-\overset{\overset{O}{\|}}{C}-OH$ E

8. Write all the resonance contributors of the benzylic cation & aniline given below (clearly Show the position of electron movement using arrows)(**0.75pt each**).

A/ 

B/